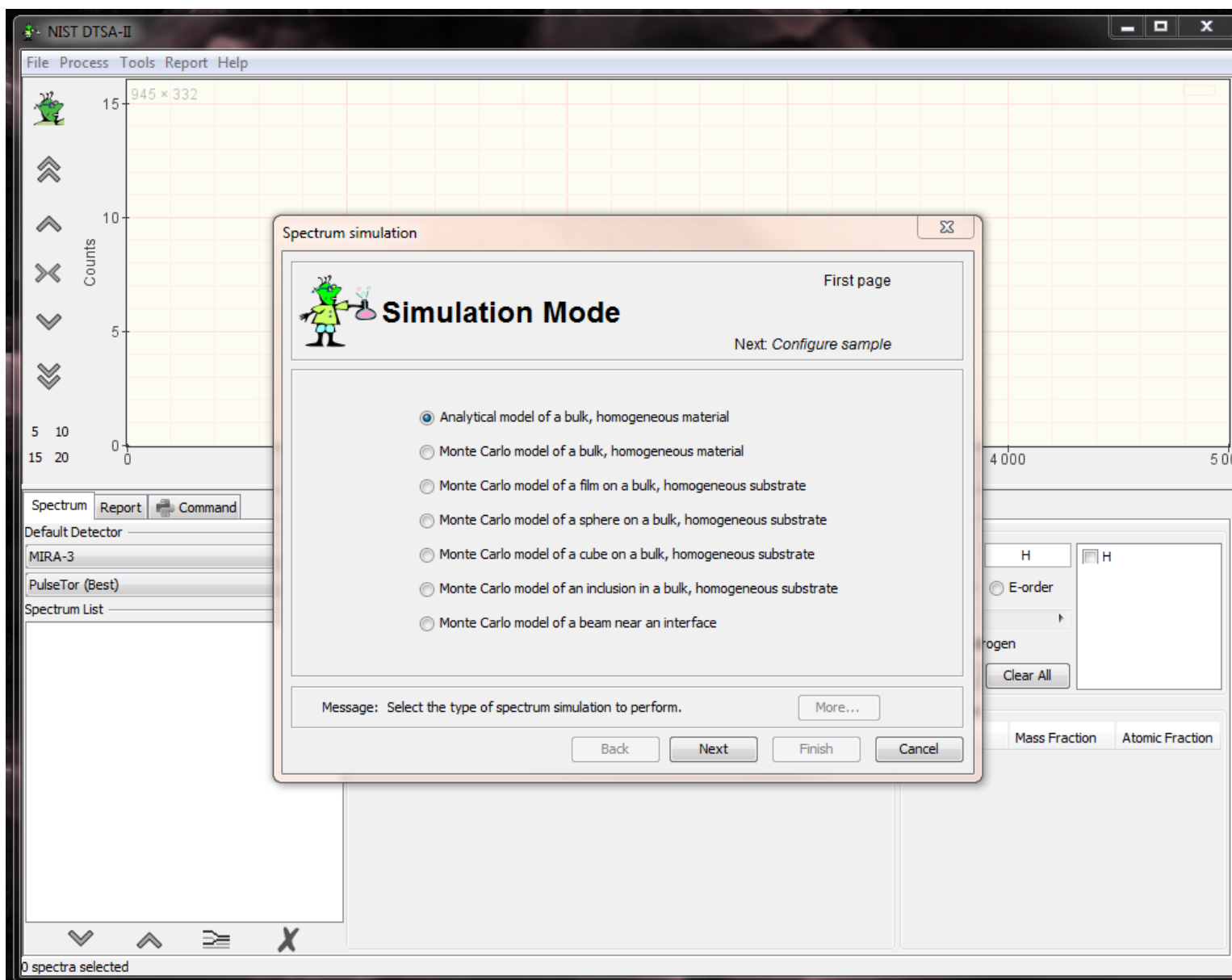
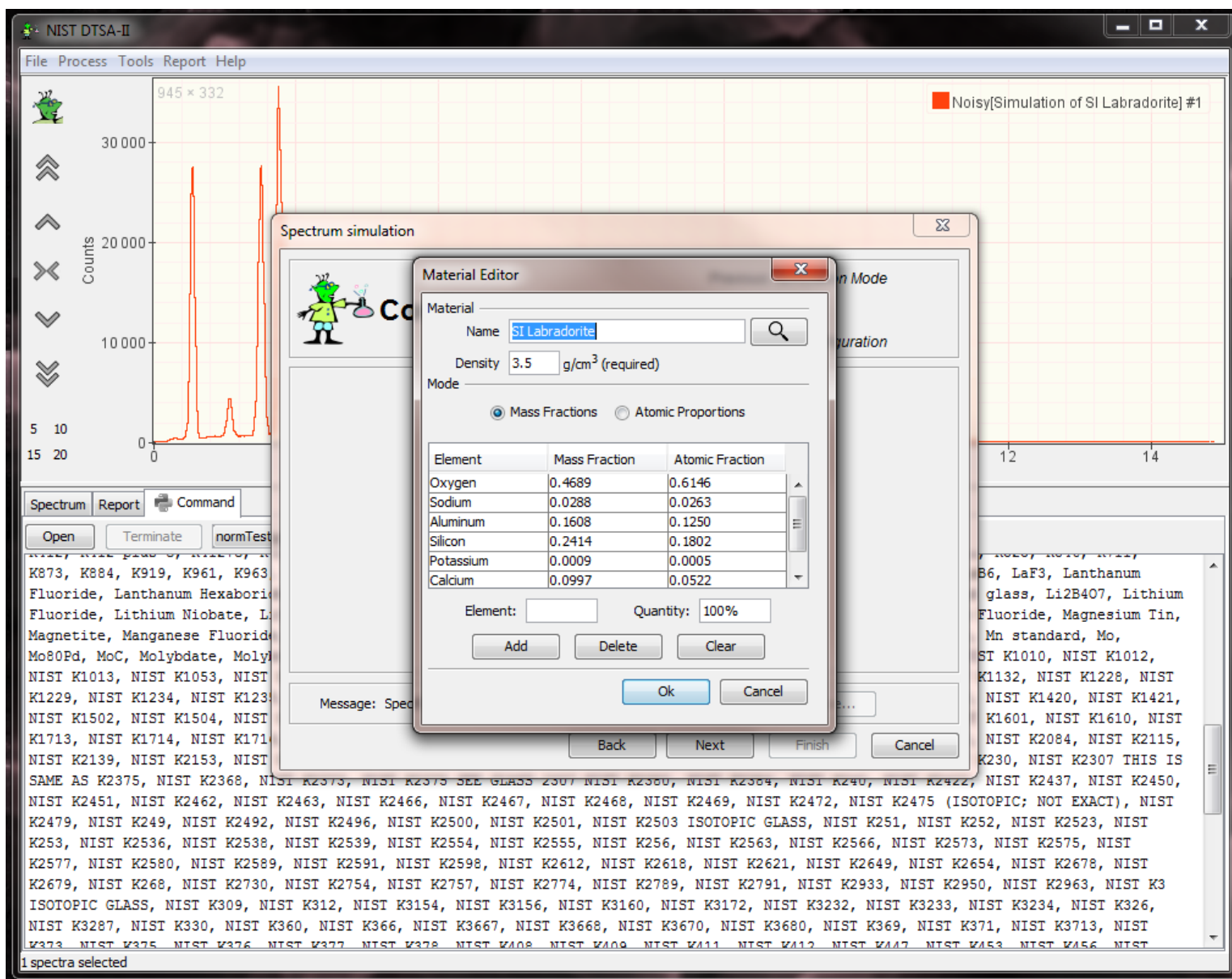
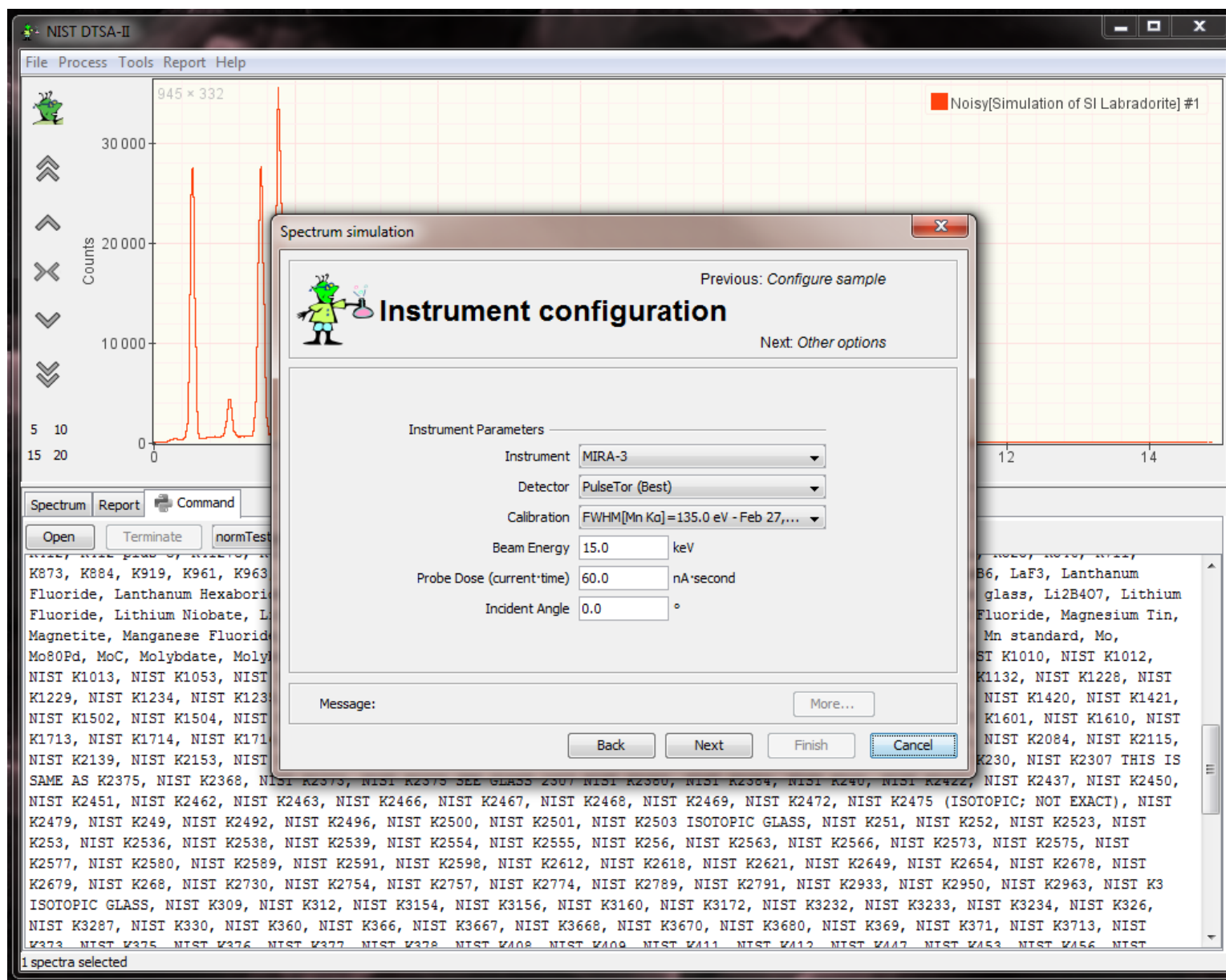


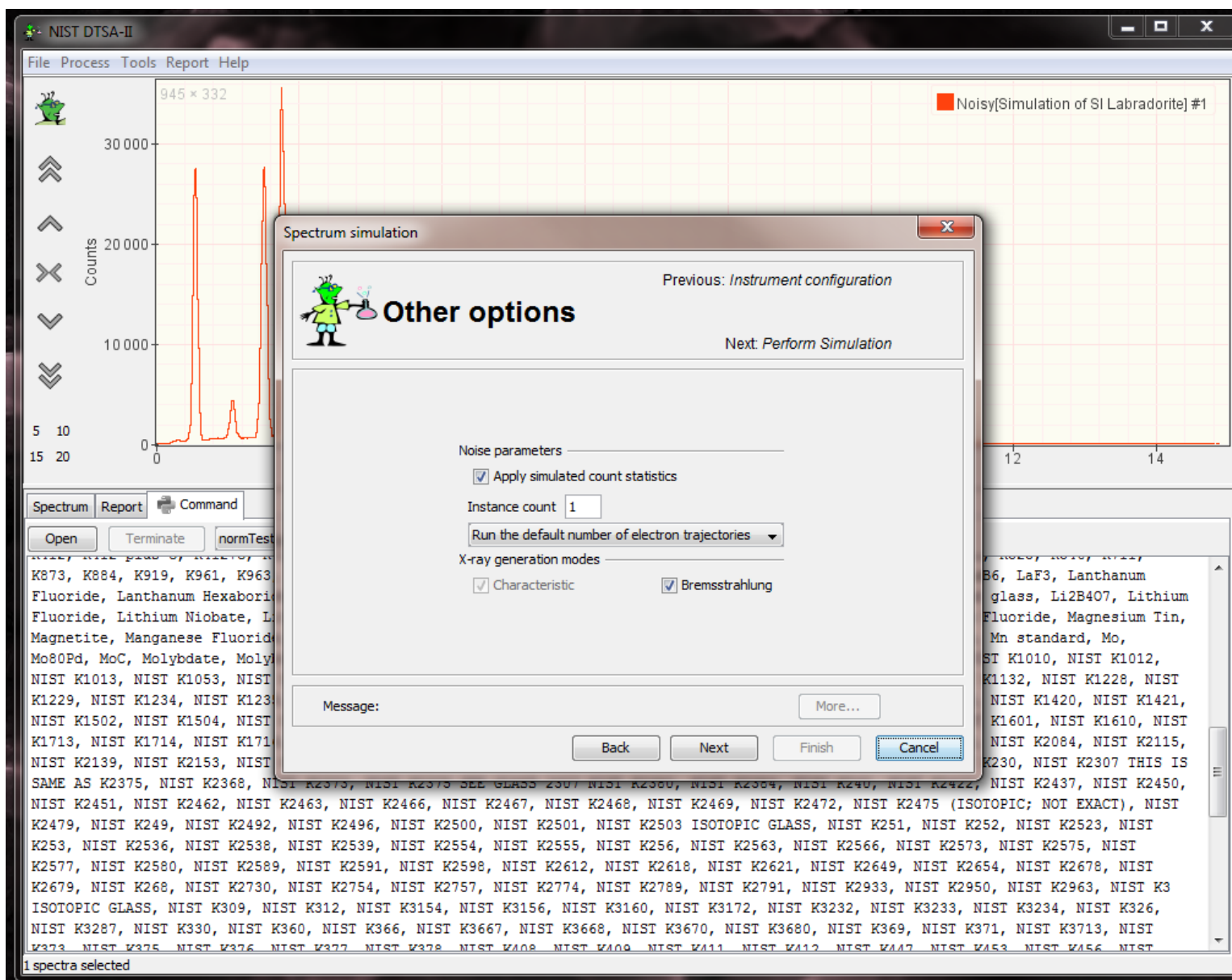
Introduction to Simulation

Analytical, Monte Carlo
Bulk and shapes

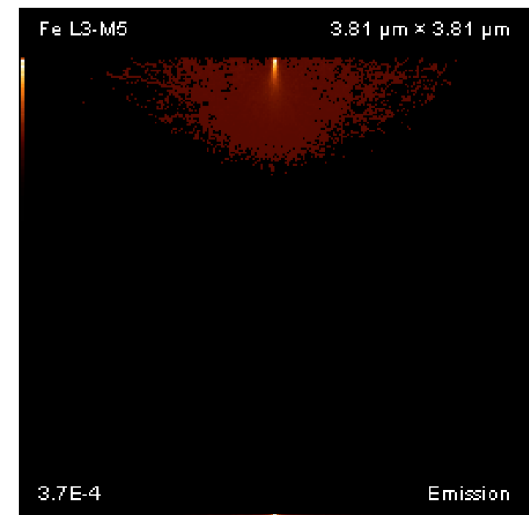
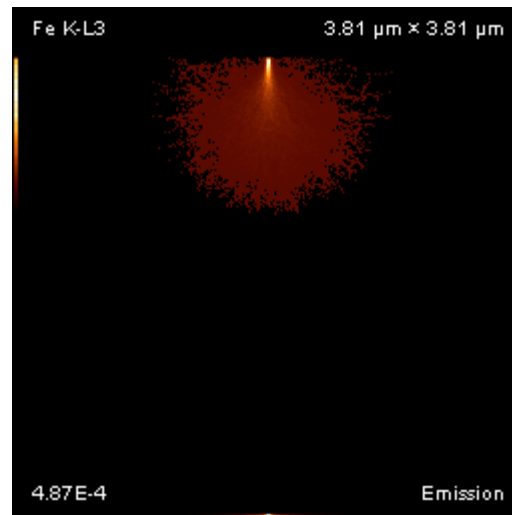
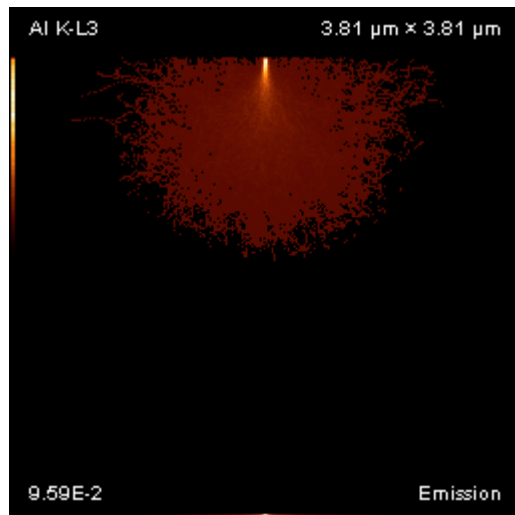
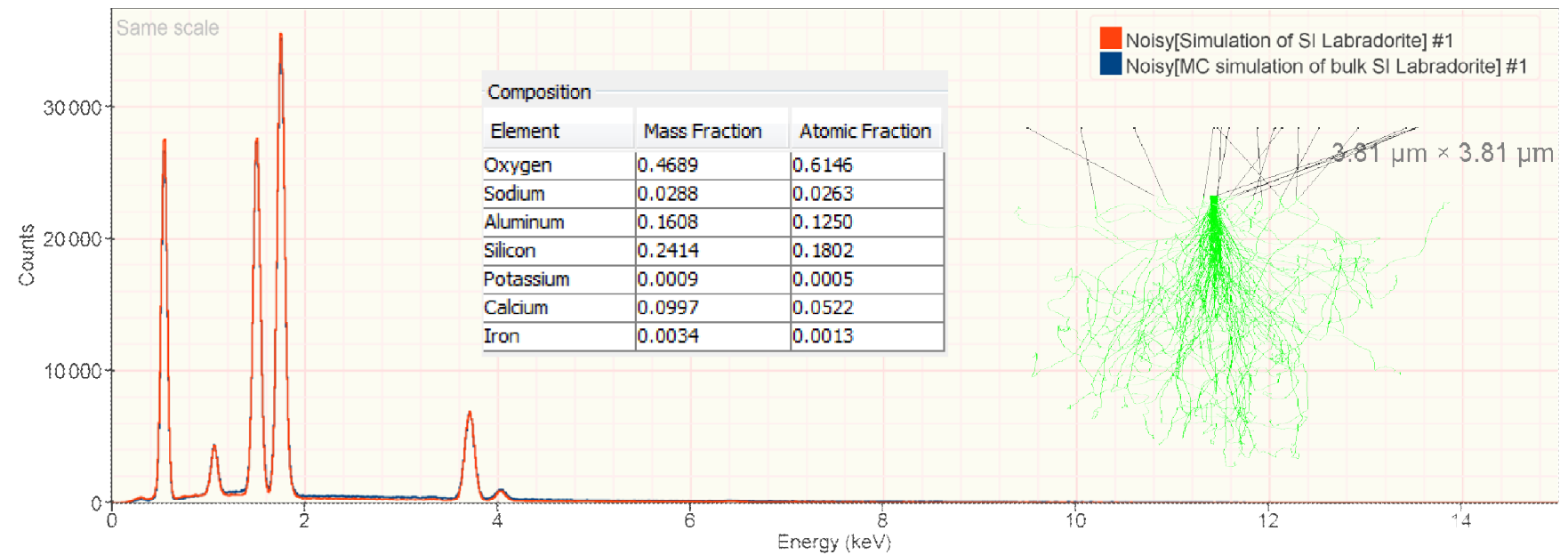




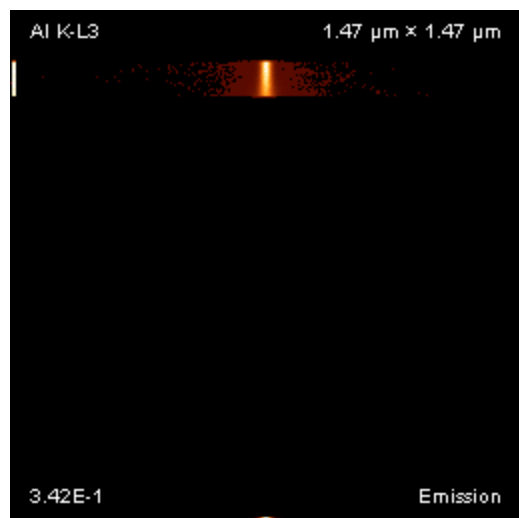




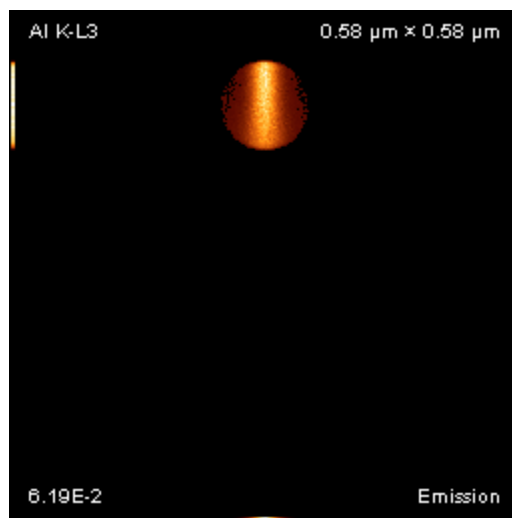
Analytical and Monte Carlo



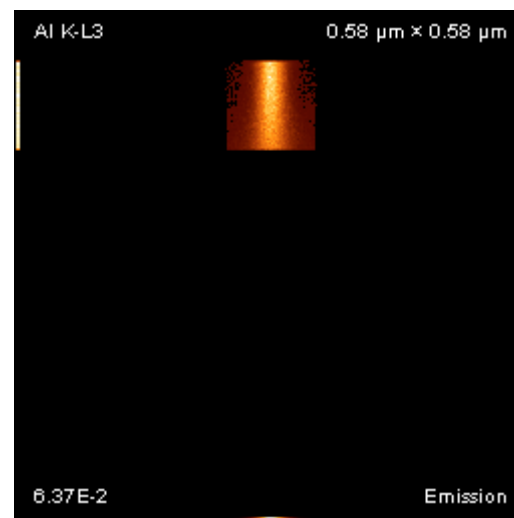
100 nm



film

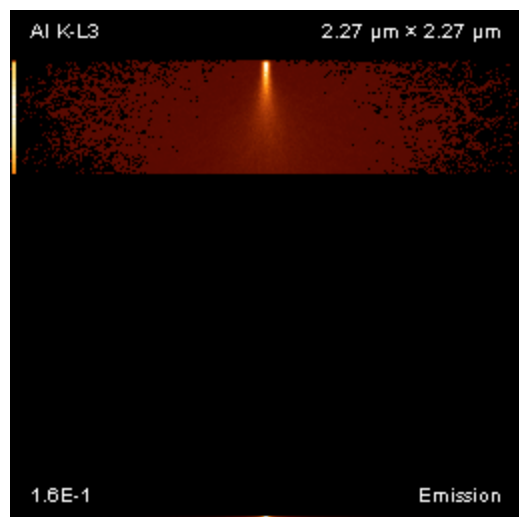


sphere

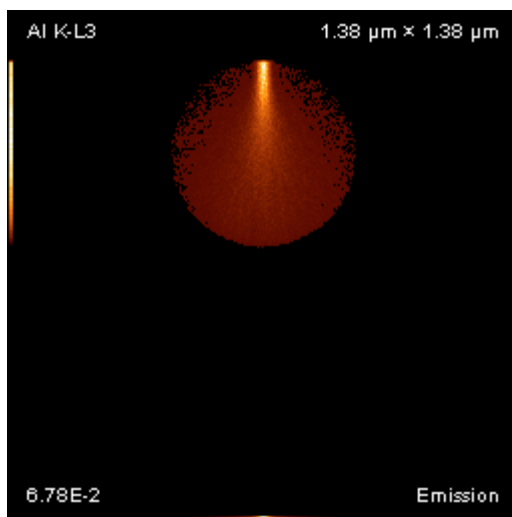


cube

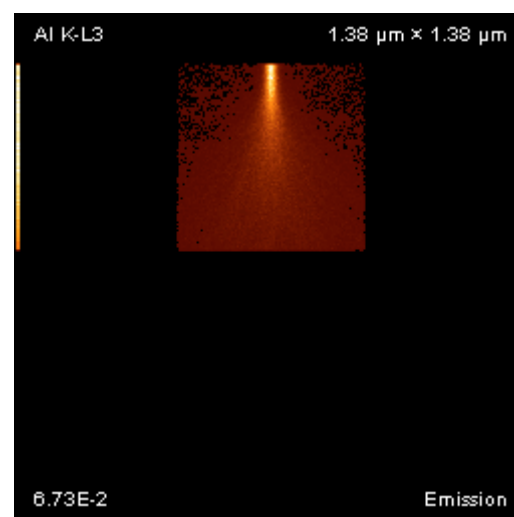
500 nm



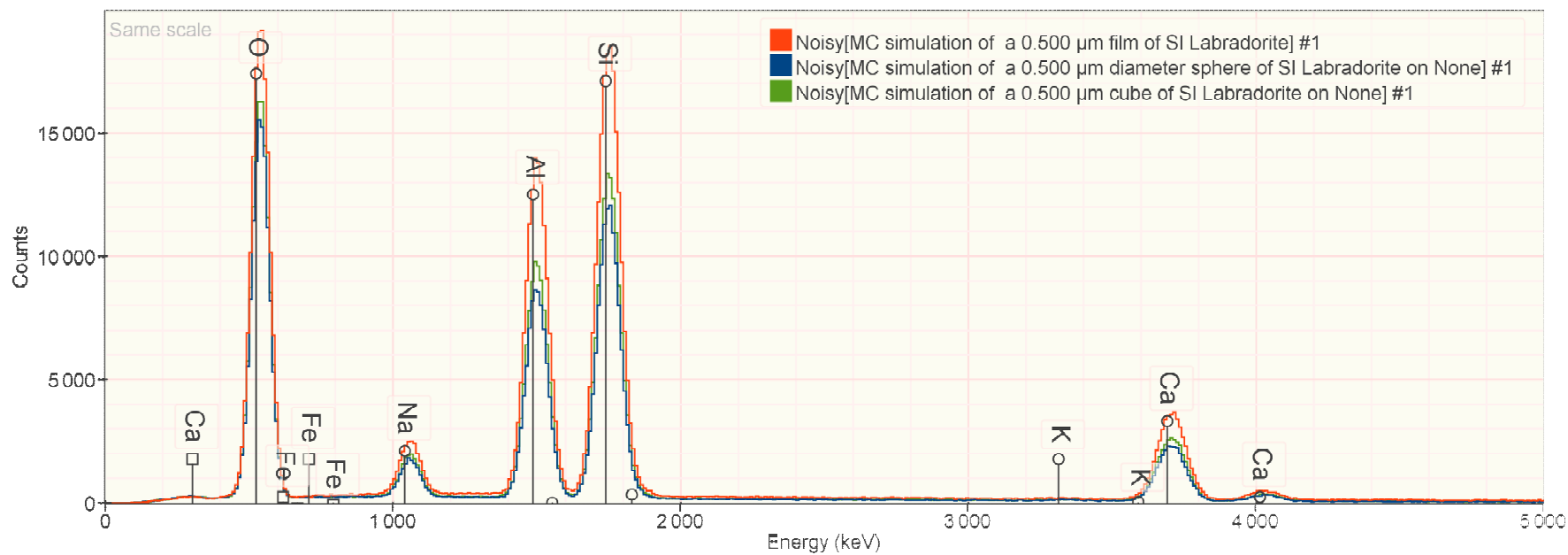
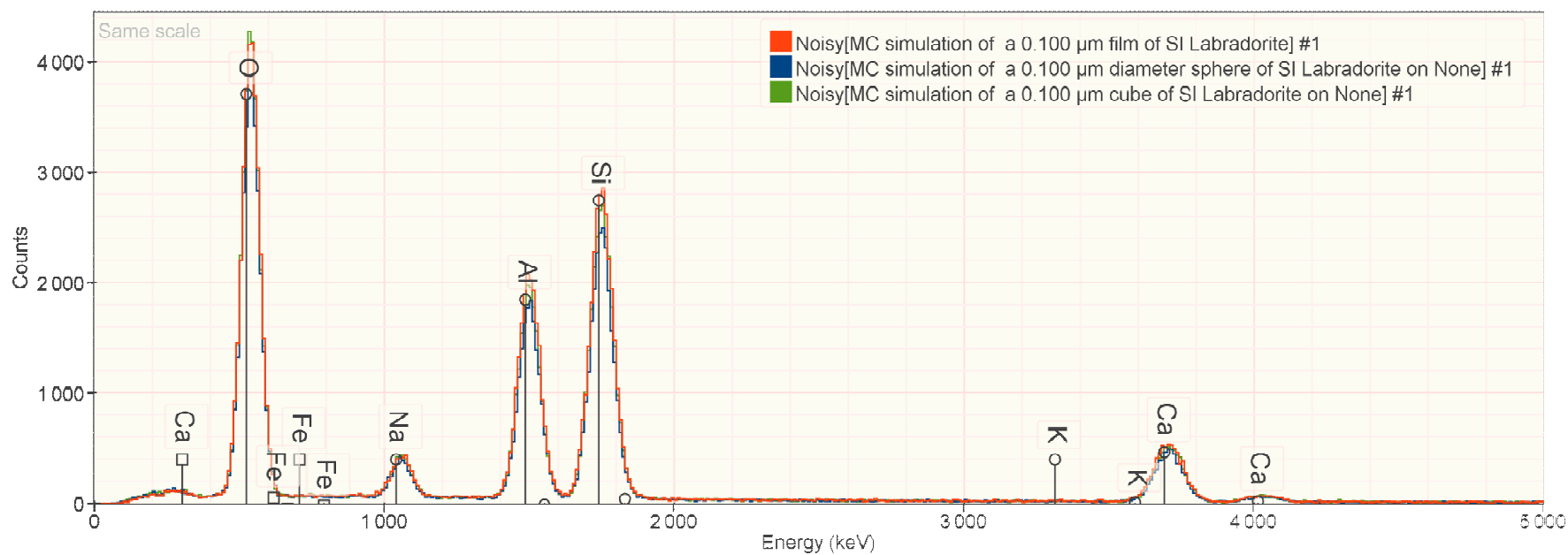
film



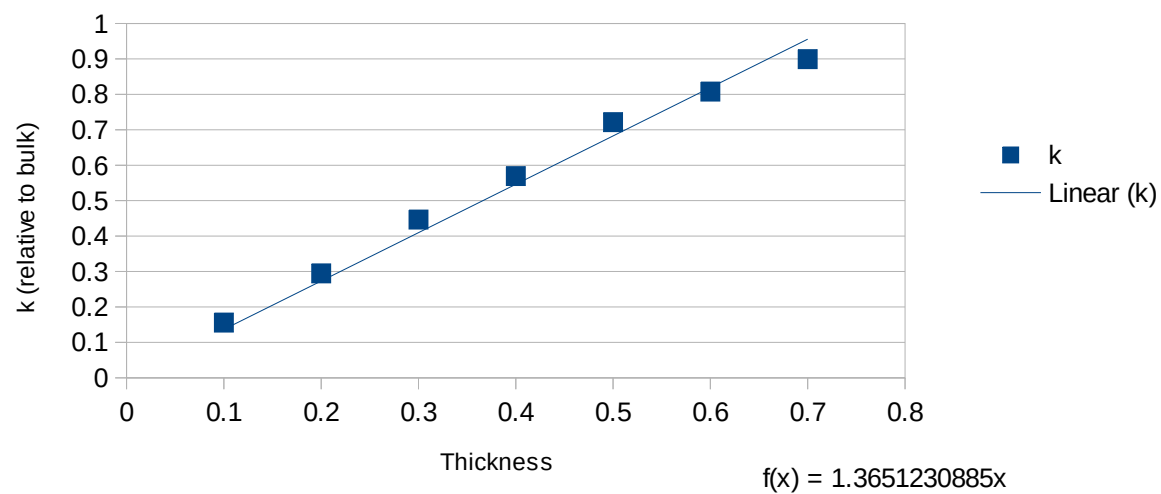
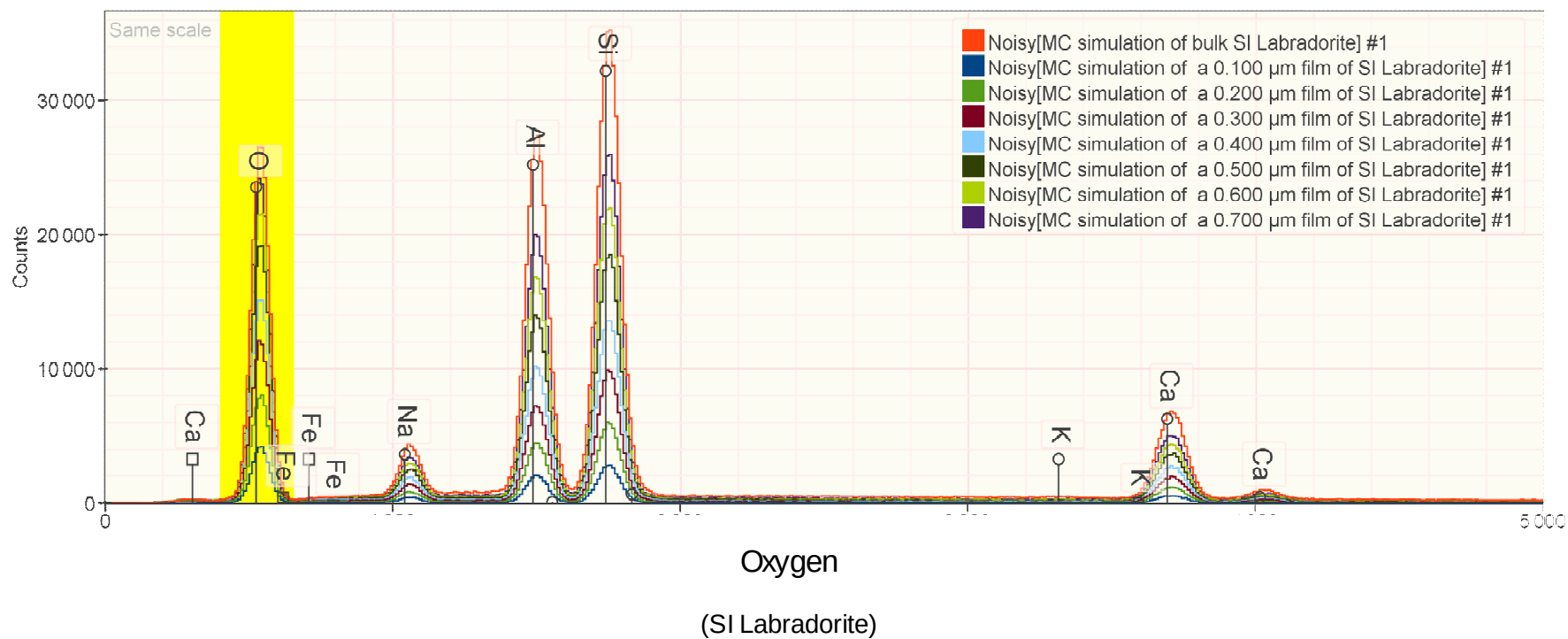
sphere



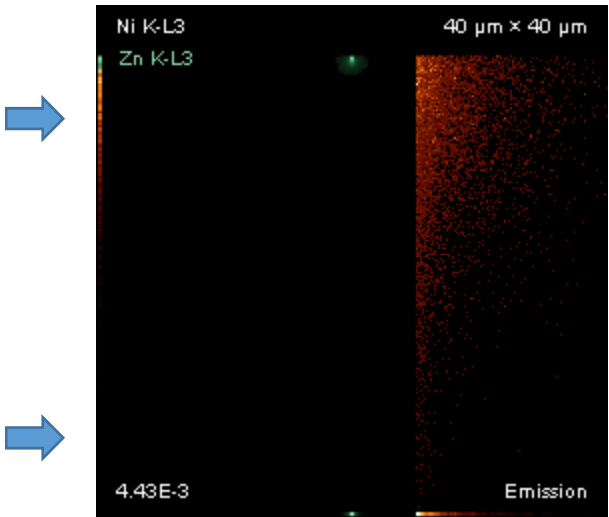
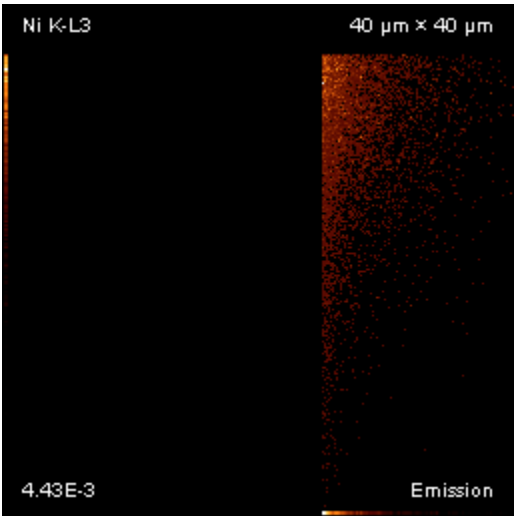
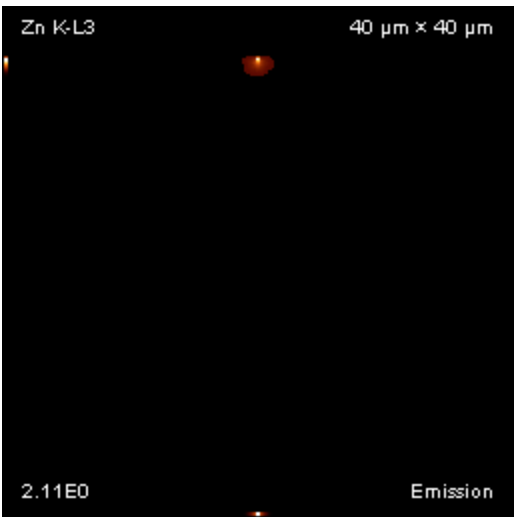
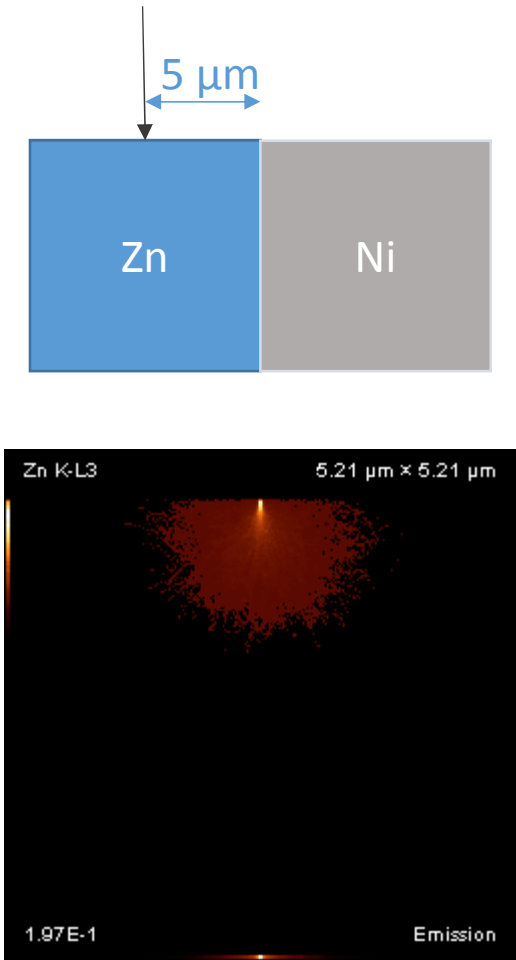
cube



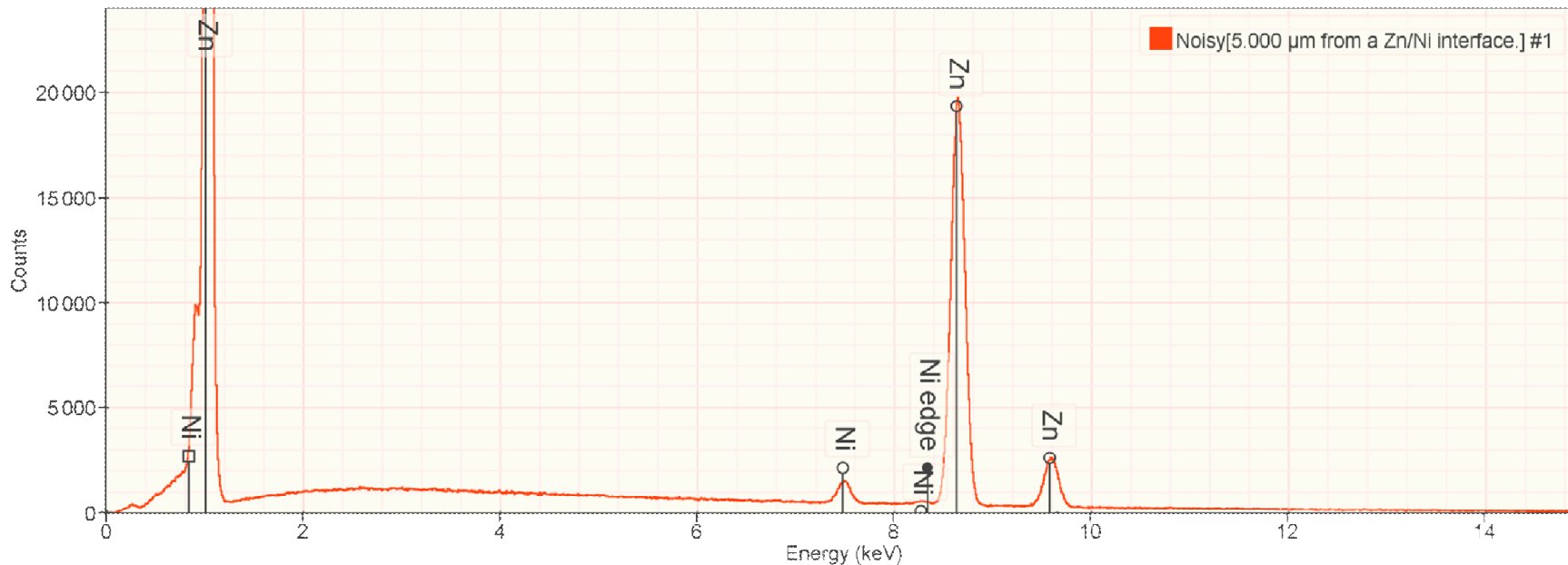
Thin film quant



Secondary fluorescence



Secondary fluorescence



Results

Spectrum	Quantity	Ni		Zn		Sum
Noisy[5.000 μm from a Zn/Ni interface.] #1 Bulk	Line Z · A · F k-ratios	Ni Kα 		Zn Kα 		
D-H = 19.946 keV	mass fraction	0.0240 ± 3.6·10 ⁻⁵ (I[std,Ni]) 0.0004(I[unk,Ni]) 9.8·10 ⁻⁶ ([μ/p]) 1.2·10 ⁻⁷ (η) [0.0004]		1.0058 ± 0.0018(I[std,Zn]) 0.0028(I[unk,Zn]) 0.0005([μ/p]) 3.3·10 ⁻⁶ (η) [0.0034]		1.0298
I = 1.000 nA	norm(mass fraction)	0.0233 ±	0.0004	0.9767 ±	0.0033	-
LT = 60.0 s	atomic fraction	0.0259 ±	0.0005	0.9741 ±	0.0033	
Residual	C:\Users\mrchie.NIST\AppData\Local\NIST\NIST DTSA-II Reports\2014\April\23-Apr-2014\residual9017401205301097157.msa					
Notes	Uncertainties are 1 σ and labeled by source. (I[std unk]: Count statistics[standard unknown], [μ/p]: Absorption correction, η: Backscatter correction, []: combined)					

What else is possible?

- Arbitrarily complex shapes
 - Plane bounded shapes
 - Spheres, hemisphere, cylinders
 - Sum and differences of shapes
- Arbitrary materials
 - $Z = 1$ to 95
 - Unlimited number of elements per material
- Scripting simulation to generate 100's or 1,000's of simulated spectra
- Simulate a variable pressure / environmental SEM